30.8 A 16nm, 1Mb, 1-to-8b-Configurable 444.21TOPS/W Fully Digital SRAM Compute-In-Memory Macro for Hybrid SNN-CNN Edge Computing

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Abstract

We present a silicon-verified fully digital SRAM-CIM macro that supports multi-bit hybrid SNN-CNN processing. Key innovations include: (1) a compact CFD-SC-RC architecture that shares SNN-CNN MAC hardware to minimize area and energy overhead while enabling adaptive model and precision selection; and (2) a PS-WR-ODM scheme that balances partial-sum storage and weight-update energy. Fabricated in 16nm FinFET CMOS, the test chip achieves 444.21TOPS/W (1bIN-8bW-14bOUT, SNN) and 62.84TOPS/W (8bIN-8bW-22bOUT, CNN).

Spiking neural networks (SNNs) [1] have demonstrated superior energy efficiency in event-driven applications such as event-based vision and audio processing, while convolutional neural networks (CNNs) [2-3] provide high accuracy in image and voice recognition tasks at the edge. Hybrid SNN-CNN models [4-6] have shown promise in balancing accuracy and compute energy, enabling robust edge inference for various applications. Meanwhile, SRAM-based digital compute-in-memory (SRAM-DCIM) [7] has been widely explored to reduce data movement and improve energy efficiency, while outperforming analog SRAM-CIM (SRAM-ACIM) [8] in MAC accuracy. Combining SRAM-DCIM with hybrid SNN-CNN models is a promising solution for diverse edge-AI tasks.

However, as shown in Fig. 30.8.1, designing SRAM-DCIM macros supporting hybrid SNN-CNN introduces two key challenges: (1) extra hardware overhead for supporting both SNN and CNN, and excessive compute energy in the digital domain; and (2) area-energy trade-off between partial sum (PSUM) storage and weight (W) update during input (IN) and W data mapping.

This work addresses these challenges with two key techniques: (1) a compact fully-digital SNN-CNN reconfigurable computing (CFD-SC-RC) architecture that shares SNN-CNN MAC circuit to reduce area and energy overhead, and supports adaptive model and precision selection across layers; and (2) a PSUM storage–weight reuse optimized data mapping (PS-WR-ODM) scheme that balances PSUM-storage minimization and weight-reuse maximization, achieving optimized register usage and weight-update energy. The test chip, fabricated in a foundry 16nm FinFET CMOS process, demonstrates 444.21 TOPS/W with a 1bIN-8bW-14bOUT configuration in SNN mode, and 62.84TOPS/W with an 8bIN-8bW-22bOUT configuration in CNN mode.

Figure 30.8.2 illustrates the proposed 1Mb CFD-SC-RC SRAM-DCIM macro, consisting of 16 64kb banks. Each bank integrates a control block, I/O block, IN/W scheduler, WL driver, 32 reconfigurable inference blocks (RIBs), an adder tree, and a digital multi-membrane potential dynamics behavior (MPDB) unit (DM-MPDBU). Each RIB contains a 2kb SRAM array, a weight stationary cell (WSC), eight cross-model dual-mode local computing cells (CMDM-LCCs), and a shift-and-add (SaA) circuit. Naïve SRAM-ACIM based hybrid SNN-CNN macros are vulnerable to process variations in MPDB functions, while prior SRAM-DCIM based hybrid SNN-CNN macros [4] support only a single MPDB type, limiting flexibility for diverse model requirements. Both architectures lack hardware sharing, incurring large area overhead. The proposed CFD-SC-RC architecture shares SNN-CNN MAC circuits to reduce area overhead, provides high tolerance to process variation, and, via the DM-MPDBU, additionally supports multiple MPDB functions beyond prior works [4], including integrate-and-fire (IF), leaky integrate-and-fire (LIF), and integer quadratic integrate-and-fire (IQIF) [9]. The IF mode provides simple spike generation; the LIF mode incorporates leakage dynamics for temporal noise filtering; and the IQIF mode introduces nonlinear integration for improved accuracy, enabling general-purpose SNN operations at the edge.

Figure 30.8.3 illustrates the CFD-SC-RC computation flow using the proposed CMDM-LCC and DM-MPDBU. Conventional bit-wise (BW)-IN multipliers (BWMs) for SNN cannot support word-wise (WW)-IN multiplication for CNN. To enable both BW-IN MAC for SNN and configurable WW-IN MAC for CNN, the proposed CMDM-LCC integrates two MUX-based multipliers (MULs), an INT8 toggle reducer (TR), dual-mode weight mapping (INT4/INT8) in the array, and an SaA circuit. Within one bank, each of eight CMDM-LCCs performs MAC operations with INs and Ws (W_{0-3}). In INT4 mode, the TR is disabled and four 4b MACs are executed per cycle: W_{0-1} map to two 4b weights (A[3:0], B[3:0]) and W_{2-3} to another two (C[3:0], D[3:0]). In INT8 mode, the TR eliminates redundant toggles, enabling two 8b MACs per cycle: W_{0-1} correspond to the four LSBs of A[7:0] and B[7:0], while W_{2-3} correspond to their four MSBs. The partial MACs ($pMAC_{0-7}$) are sent directly to the adder tree in SNN mode, while in CNN mode $pMAC_{0-7}$ are passed to the SaA circuit for WW-accumulation before entering the adder tree. Meanwhile, conventional MPDB blocks only support a single MPDB function, limiting flexibility for diverse SNN models. The proposed DM-MPDBU supports three functions: (1) IF mode—membrane potential (MP) accumulates until reaching threshold (peak), then fires and resets; (2) LIF mode—MP accumulates with leakage subtraction before proceeding as IF; and (3) IQIF mode—MP accumulates with an integer quadratic (IQ) term dynamically defined by parameters (pdeth, rest, thresh, α, β): if MP[t] < pdeth, IQ=(α×rest–MP[t])) >> 3; otherwise, IQ=(β×(MP[t]–thresh)) >> 3.

Figure 30.8.4 illustrates the proposed PS-WR-ODM scheme. In conventional timestep (TS)-first data mapping, all input neurons (IN_{0-K}) are accessed sequentially within each TS, minimizing PSUM storage but incurring high weight-update energy and long latency because weight reuse cannot be leveraged for early MPDB operations. In contrast, prior weight-reuse data mapping [10] accesses same-index INs (e.g., IN_0) across all TSs (TS_{0-N}), maximizing weight reuse and throughput but requiring large PSUM storage. The proposed PS-WR-ODM adopts a hybrid strategy in which TSs are grouped into blocks with user-defined size (e.g., pairs of adjacent TSs), and weight-reuse mapping is applied within each block. This approach achieves balanced performance in weight-update energy and latency with only marginal PSUM buffer overhead. For example, as shown in the timing diagrams, the conventional TS-first data mapping incurs long weight-load cycles, while the previous weight-reuse mapping demands excessive PSUM buffering. The proposed PS-WR-ODM achieves comparable latency reduction and energy savings without significant PSUM overhead.

Figure 30.8.5 shows the simulated performance of the proposed schemes. The CFD-SC-RC reduces macro area by 33% and improves energy efficiency by 2.31× over prior hybrid SNN-CNN CIM design [4]. The PS-WR-ODM achieves >1.22× improvement in FoM1, defined as: energy efficiency/(area×latency), compared to conventional timestep-first and weight-reuse-first [10] mappings. The combined implementation of CFD-SC-RC, CMDM-LCC, and PS-WR-ODM further delivers an FoM2, defined as inference accuracy×energy efficiency/(area×latency)], of 5.48× over baseline designs [4,10].

Figure 30.8.6 presents the measurement results of the 16nm SRAM-DCIM macro and the position chart versus prior silicon-verified works. The Shmoo plot confirms 1.3ns computing latency (T_{MAC}) at 0.8V with an 8bIN-8bW-22bOUT configuration. The proposed macro achieves up to 8.2× improvement in FoM3, defined as: IN precision×W precision×energy efficiency, over prior works. Figure 30.8.7 presents the die micrograph and chip performance summary. To the best of our knowledge, this is this chip is the first silicon-verified fully digital SRAM-CIM macro that supports multi-bit hybrid SNN–CNN processing.

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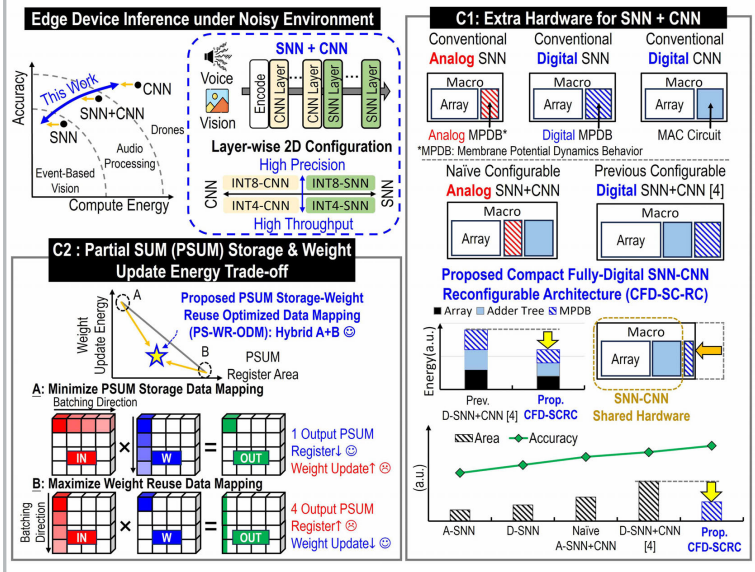


Figure 30.8.1: Motivation and challenges for hybrid SNN-CNN SRAM-CIM.

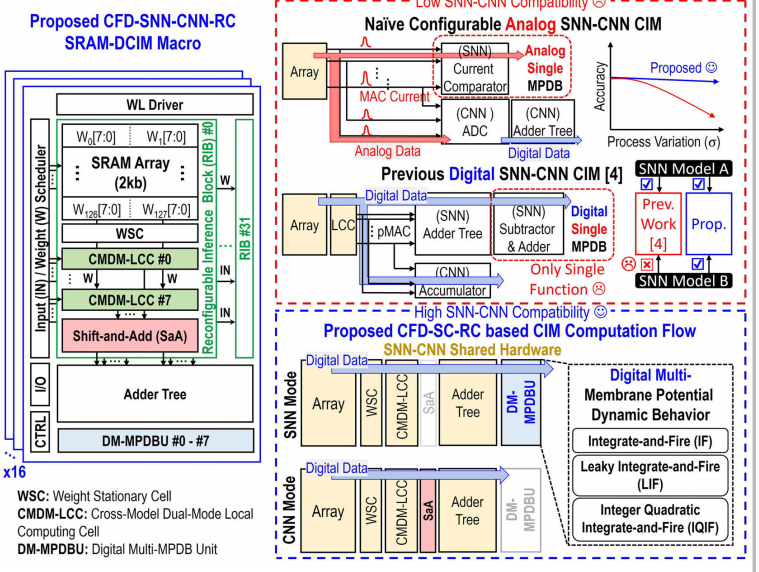


Figure 30.8.2: Structure of proposed SRAM-DCIM macro and CFD-SC-RC operation.

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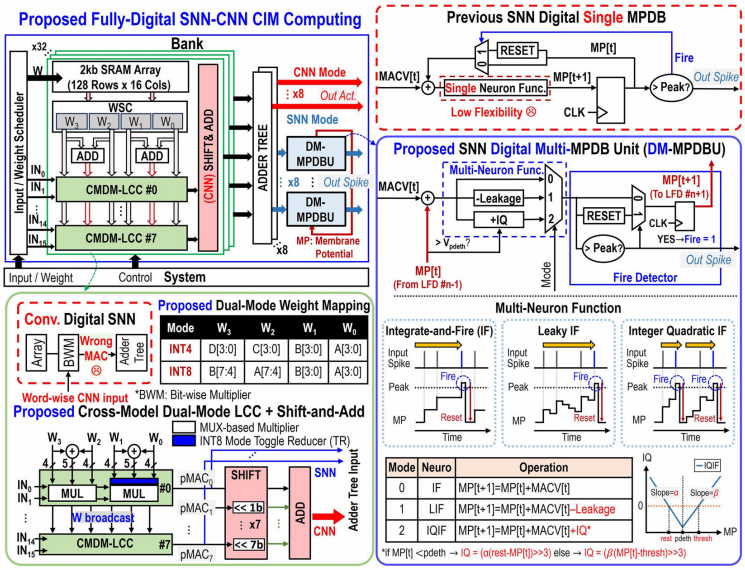


Figure 30.8.3: CMDM-LCC and DM-MPDBU structure and operation.

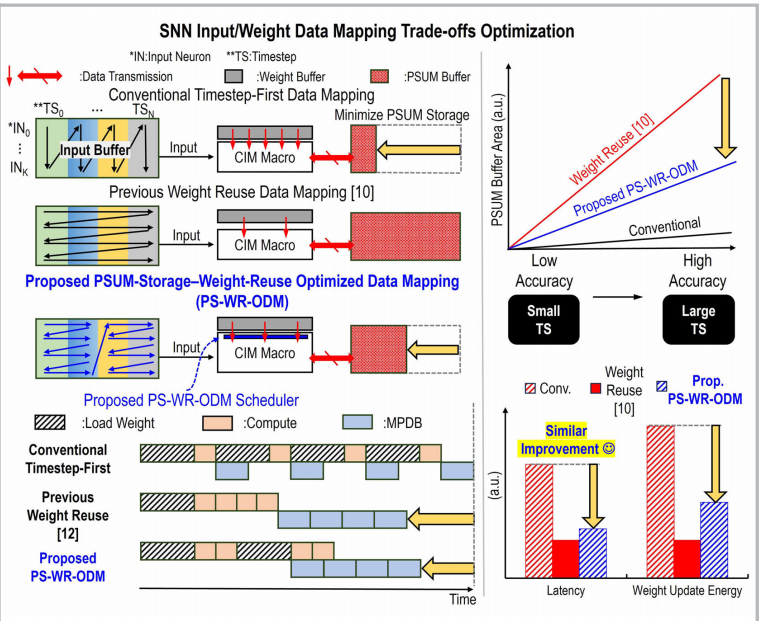


Figure 30.8.4: PS-WR-ODM operational concept.

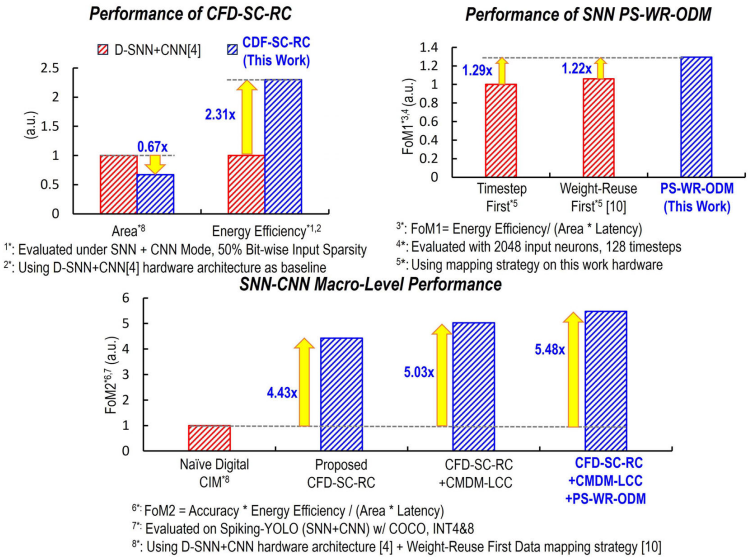


Figure 30.8.5: Simulated performance of proposed schemes.

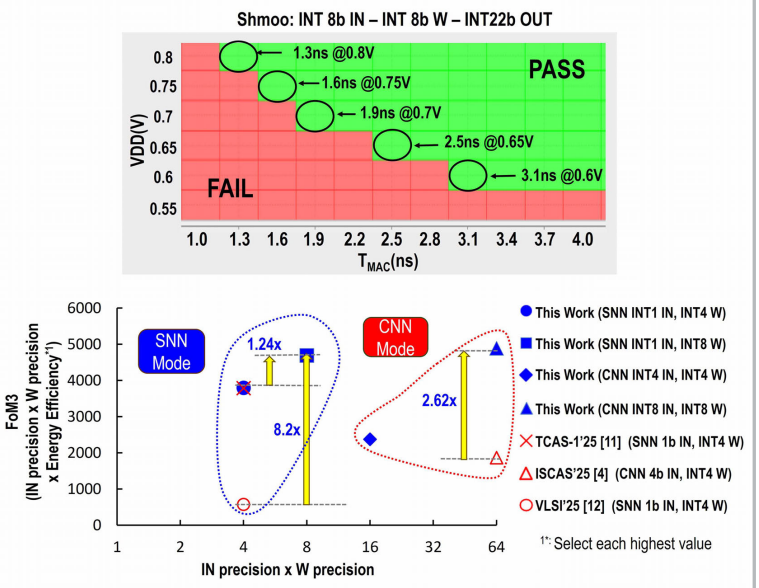
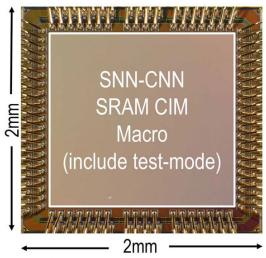


Figure 30.8.6: Measurement results and position chart.



CHIP SUMMARY				
Technology	16nm FinFET CMOS			
Domain	Digital			
Cell Type	Compact 6T-SRAM			
Macro size	1Mb			
Mode	SNN		CNN	
Input precision (bit)	1		4	8
Weight precision (bit)	4	8	4	8
Number of input channels	128	64	128	64
Number of output channels	128		16	
Output precision (bit)	11	14	15	22
Supply voltage (V)	0.6 - 0.8			
Access time*1 (ns)	1.3			
Throughput*1,2 (TOPS)	25.21	12.60	3.15	1.58
Energy Efficiency*2 (TOPS/W) (Average Performance*4)	870.28	444.21	123.62	62.84
Energy Efficiency*2 (TOPS/W) (Peak Performance*3)	949.41	586.36	148.35	76.35
Area Efficiency*1,2 (TOPS/mm^2)	1.69 - 26.99			
SNN MPDB Type	LIF / IF / IQIF			
Accuracy Loss	No			
Silicon-Verified	Yes			

1*: MAC computing @VDD = 0.8V
2*: One operation (OP) represents one multiplication or one addition
3*: Measured @VDD = 0.8V, 50% input sparsity
4*: Measured @VDD = 0.6V, 90% input sparsity

Figure 30.8.7: Die micrograph and performance summary table.

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